

MOHD. HAMZA ANSARI

Python Developer | Generative AI, RAG & Machine Learning

Ayodhya, Uttar Pradesh, India • +91-7275418812 • mohdhamza0456@gmail.com

LinkedIn: www.linkedin.com/in/mohd-hamza-648a15319

PROFESSIONAL SUMMARY

MCA graduate (2026) and Python Developer specializing in Generative AI, Retrieval-Augmented Generation (RAG), and machine learning. Hands-on experience building LLM-powered systems with SBERT embeddings, semantic search, and vector retrieval using Python, PyTorch, Scikit-learn, and Streamlit. Strong foundation in data structures, algorithms, OOP, and SQL, with proven ability to take projects end-to-end — from data preprocessing and model development through RAG pipeline design to deployment. Seeking an entry-level Python / Generative AI Developer role to build scalable, real-world AI solutions.

TECHNICAL SKILLS

Programming Languages: Python, SQL, C, C++

Generative AI & LLMs: Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Prompt Engineering, LLM API Integration, Semantic Search, Vector Embeddings (SBERT), Vector Databases

Machine Learning: Supervised & Unsupervised Learning (Linear Regression, Random Forest, K-Means), Feature Engineering, Model Evaluation & Optimization, Data Preprocessing

Deep Learning & NLP: PyTorch, Natural Language Processing (NLP), SBERT Embeddings, Semantic Text Similarity

Data Analysis & Visualization: Data Cleaning, Statistical Analysis, Pandas, NumPy, Matplotlib

Web & Core CS: Object-Oriented Programming (OOP), Data Structures & Algorithms, HTML, CSS, PHP

Tools & Deployment: Scikit-learn, Streamlit, Git, REST API Integration, Model Deployment

PROJECTS

AI Resume Analyzer & Optimizer — RAG-Based Resume Intelligence Tool

- Built a Retrieval-Augmented Generation (RAG) system combining SBERT semantic embeddings with a large language model to analyze resumes against job descriptions and automate ATS-style match scoring.
- Designed the retrieval pipeline to embed resume and job-description content, compute semantic similarity, and feed the most relevant context to the LLM for grounded, role-specific feedback.
- Integrated a generative AI / LLM layer to produce keyword recommendations and personalized, context-aware resume improvement suggestions.
- Developed an interactive Streamlit interface for real-time scoring, job-description matching, and downloadable feedback reports.

Tech stack: Python, RAG, SBERT, LLM API, Scikit-learn, Streamlit, Git

DriveScore — ML-Driven Car Rating & Performance Analyzer

- Developed a machine learning application that predicts and rates cars from features such as brand, model, year, mileage, and price using a Random Forest model.
- Implemented end-to-end data preprocessing, feature selection, and model evaluation pipelines to deliver reliable, data-backed ratings.
- Translated model outputs into clear, interpretable ratings that help users make informed car-purchase decisions.

Tech stack: Python, Scikit-learn, Pandas, NumPy, Matplotlib

EDUCATION

Master of Computer Applications (MCA)

2024 – 2026

Jaypee Institute of Information Technology, Noida, Uttar Pradesh

Bachelor of Science (B.Sc.) — Mathematics

2020 – 2023

Dr. RML Avadh University, Ayodhya, Uttar Pradesh

RELEVANT COURSEWORK

Data Structures & Algorithms · Machine Learning · Deep Learning · Natural Language Processing · Database Management Systems · Object-Oriented Programming · Operating Systems

ADDITIONAL INFORMATION

- Strengths:** Strong analytical problem-solver who ships end-to-end AI/ML projects independently.
- Interests:** Applying Generative AI, RAG, and ML to real-world and sustainable-technology problems.